

| Rev | Date       | Eng | Changes                                          |
|-----|------------|-----|--------------------------------------------------|
| AA  | 2026-10-07 | NB  | First release, derived from Sofar mote 000639-AB |

# nereus\_Pi\_shield\_ADIN2111

Variant: Standard

P/N: PCB-000001-AA

Index

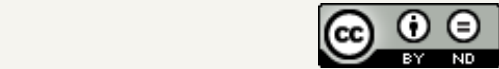
- 1 Cover
- 2 Top-Level
- 3 ADIN2111 (Single-Pair Ethernet)
- 4 Load Switch (payload port)
- 5 PoDL (Power over Data Line)
- 6 Bus Power Monitor
- 7 Power (buck converters)

Pi shield renders: TBD (after layout)

I2c Address Table

| Device        | I2C Address |
|---------------|-------------|
| Power Monitor | 0x41        |

**DIMENSIONS:** TBD (after layout)

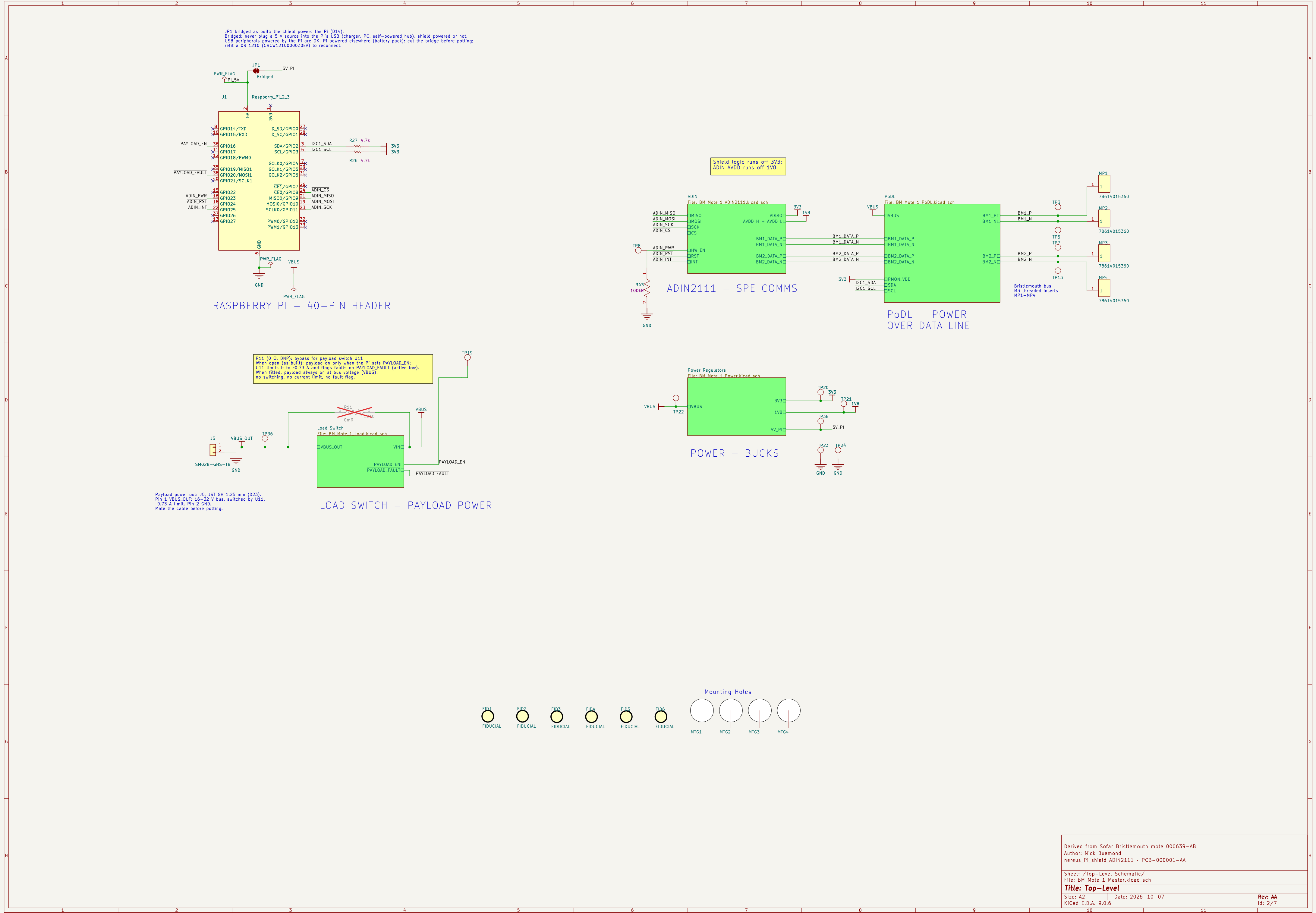


Top-Level Schematic  
File: BM Mote 1 Master.kicad\_sch

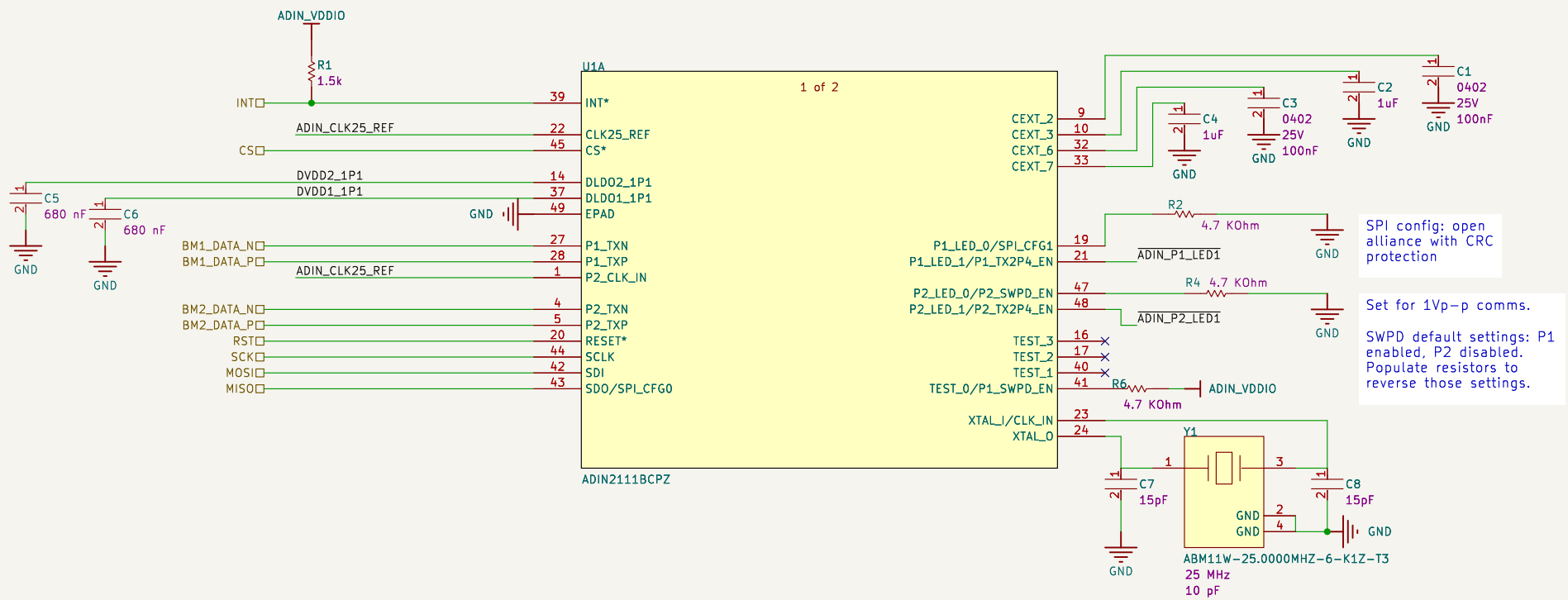


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|                                                                                                                     |                  |         |         |
|---------------------------------------------------------------------------------------------------------------------|------------------|---------|---------|
| Derived from Sofar Bristlemouth mote 000639-AB<br>Author: Nick Buemond<br>nereus_Pi_shield_ADIN2111 · PCB-000001-AA |                  |         |         |
| Sheet: /<br>File: nereus_Pi_shield_ADIN2111.kicad_sch                                                               |                  |         |         |
| <b>Title: Cover Page</b>                                                                                            |                  |         |         |
| Size: A3                                                                                                            | Date: 2026-10-07 |         | Rev: AA |
| KiCad E.D.A. 9.0.6                                                                                                  |                  | Id: 1/7 |         |

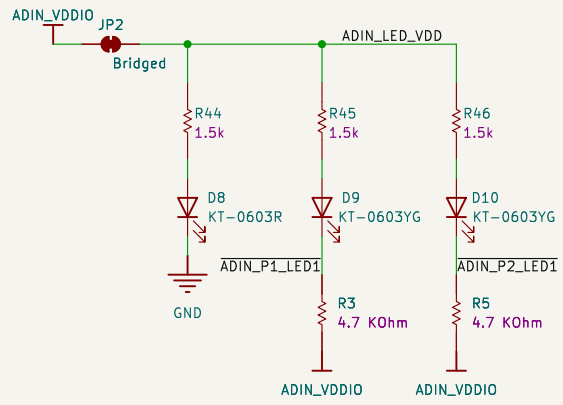
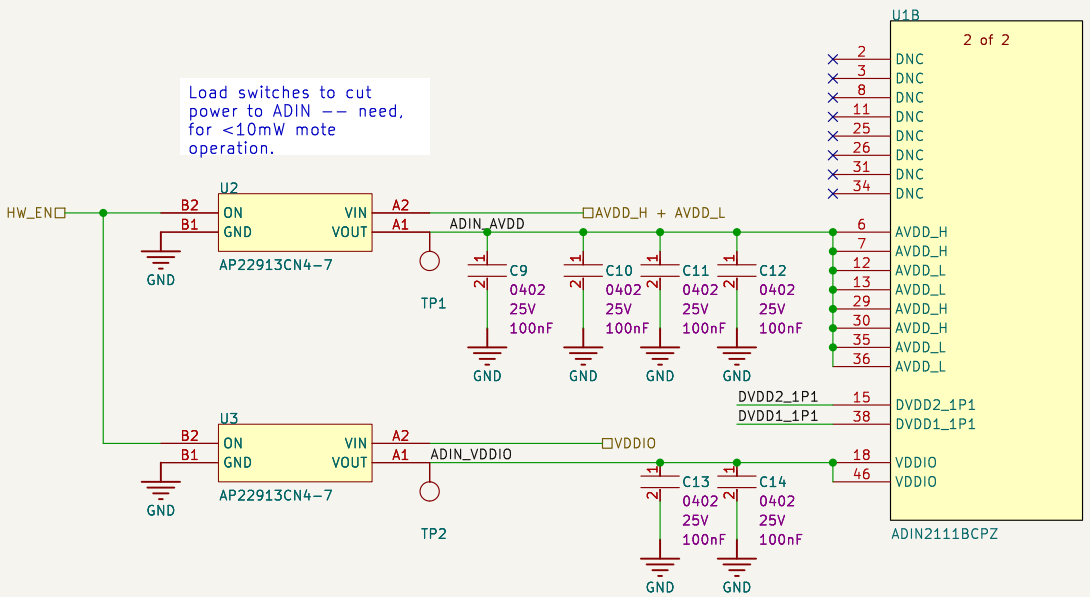


ADIN2111 Single-Pair Ethernet



ADIN2111 status LEDs: power, port 1 / port 2 link-activity

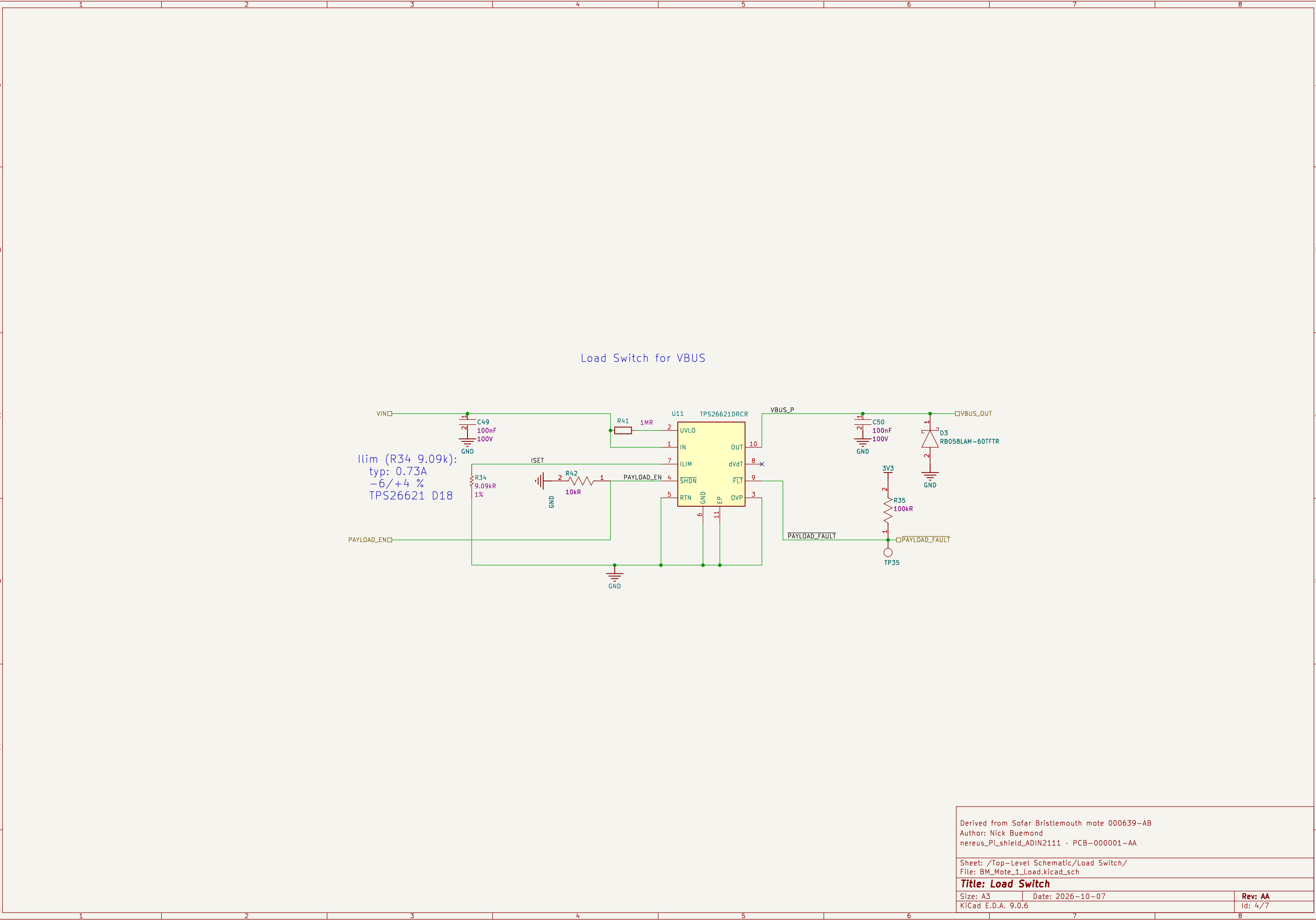
ADIN2111 supplies: U2 (AVDD) and U3 (VDDIO) switched on by ADIN\_PWR



Derived from Sofar Bristlemouth mote 000639-AB  
Author: Nick Buemond  
nereus\_Pi\_shield\_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/ADIN/  
File: BM\_Mote\_1\_ADIN2111.kicad\_sch

| Title: ADIN2111    |                  |         |
|--------------------|------------------|---------|
| Size: A3           | Date: 2026-10-07 | Rev: AA |
| KiCad E.D.A. 9.0.6 |                  | Id: 3/7 |



Derived from Sofar Bristlemouth mote 000639-AB  
Author: Nick Buemond  
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Sheet: /Top-Level Schematic/Load Switch/  
File: BM\_Mote\_1\_Load.kicad\_sch

**Title: Load Switch**

Size: A3  
KiCad E.D.A. 9.0.6

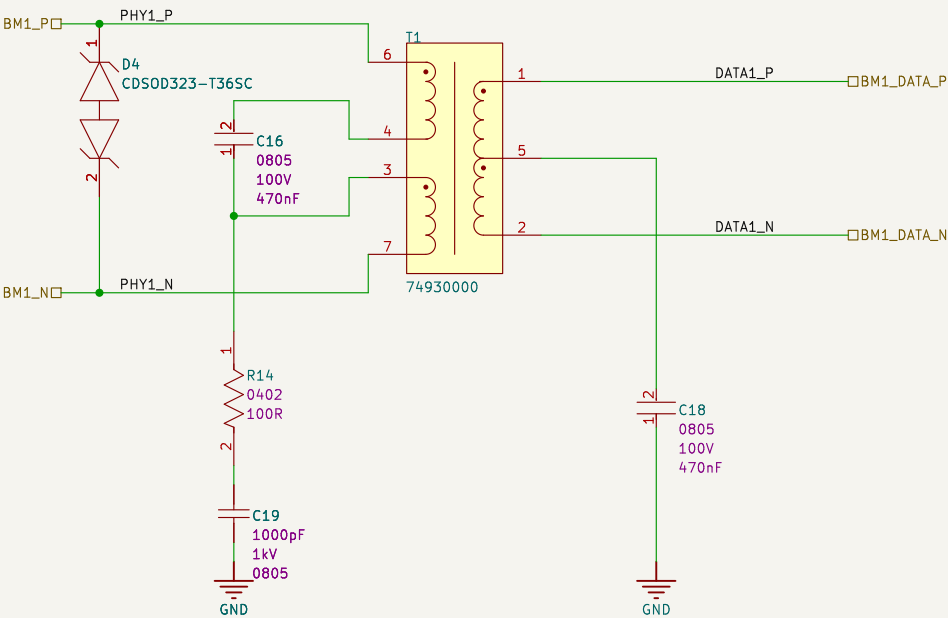
Date: 2026-10-07

Rev: AA

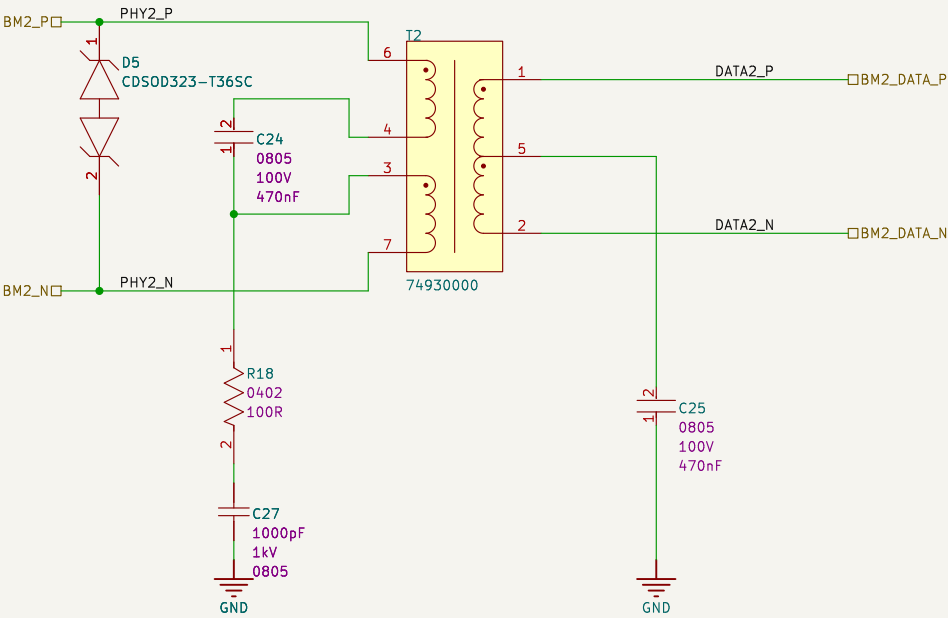
Id: 4/7

Power Over Dataline

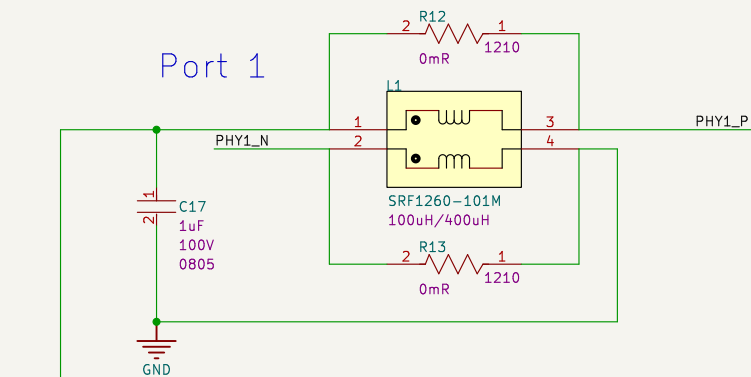
Coupling Network – Port 1



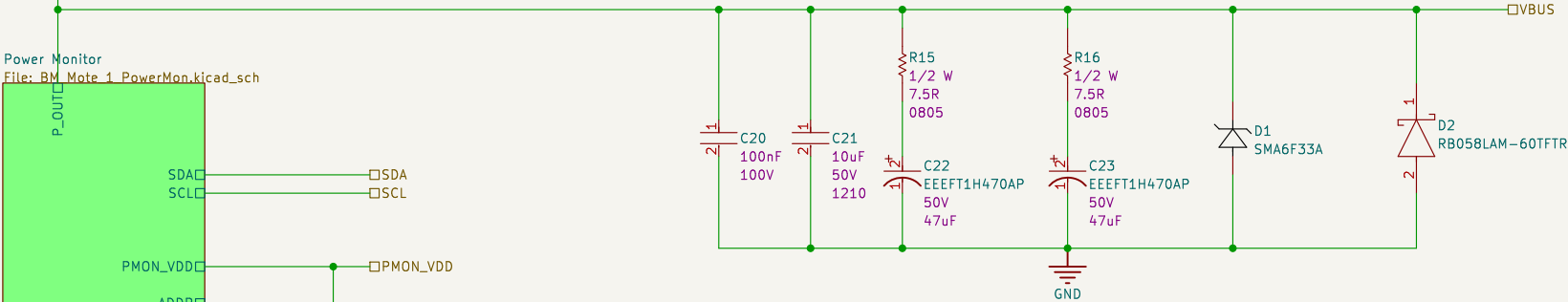
Coupling Network – Port 2



Port 1

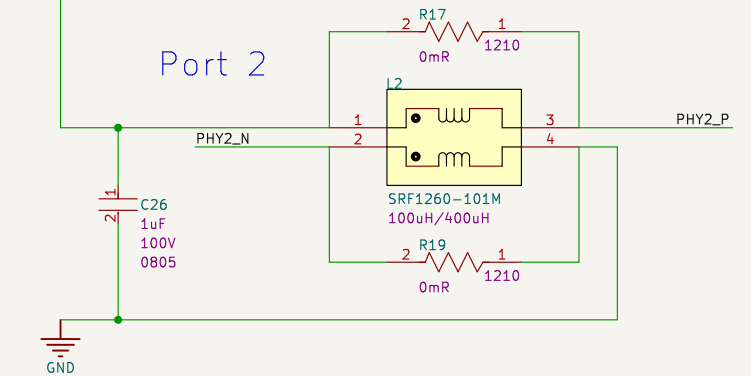


DC Connection + Circuit protections



Snubber resistors must be at least 1/2 W, to support hot-plugging.

Port 2



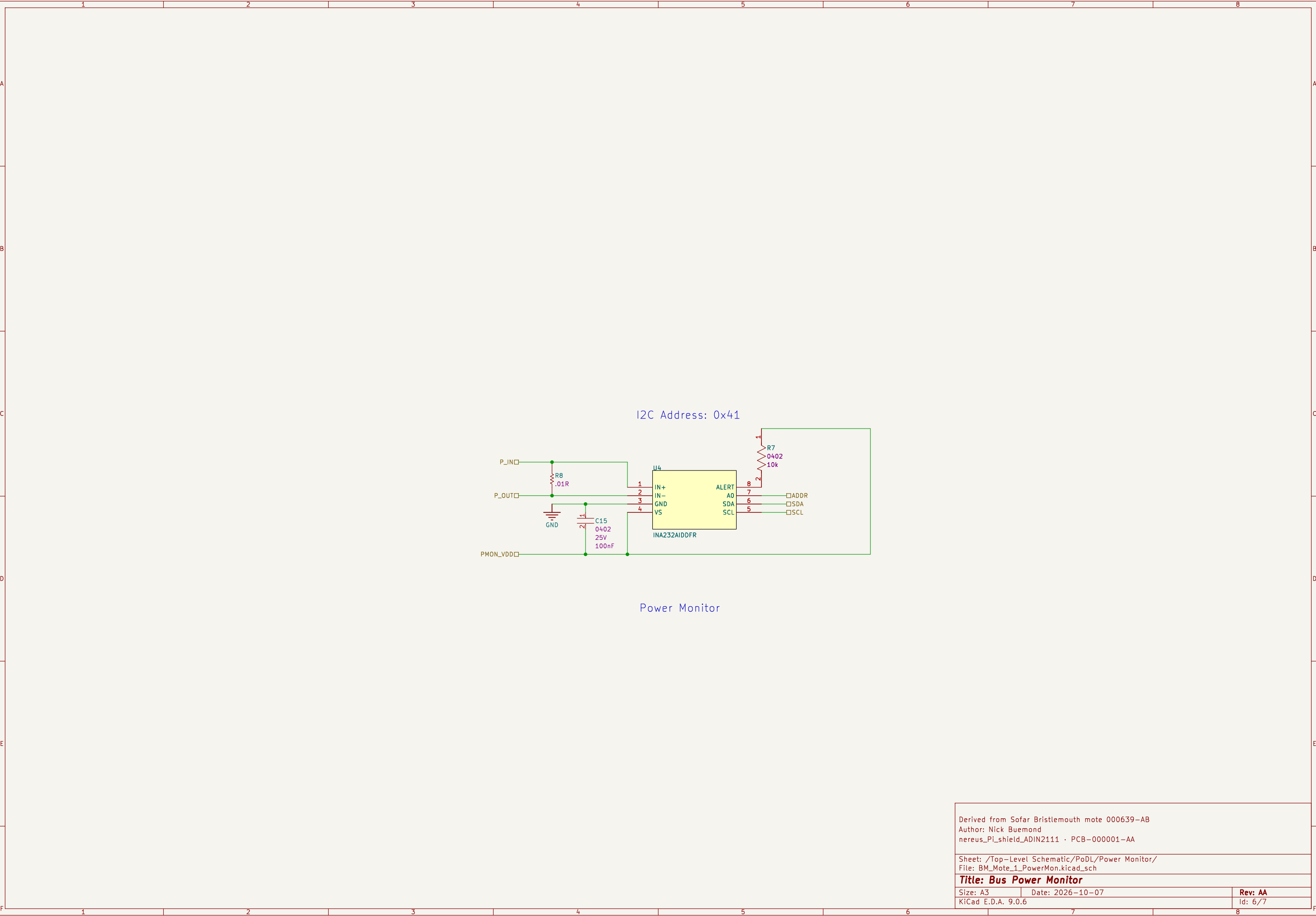
Derived from Sofar Bristlemouth mote 000639-AB  
Author: Nick Buemond  
nereus\_Pi\_shield\_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/  
File: BM\_Mote\_1\_PoDL.kicad\_sch

**Title: PoDL (Power over Data Line)**

Size: A3  
Date: 2026-10-07  
KiCad E.D.A. 9.0.6

Rev: AA  
Id: 5/7



Derived from Sofar Bristlemouth mote 000639-AB  
Author: Nick Buemond  
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Sheet: /Top-Level Schematic/PoDL/Power Monitor/  
File: BM\_Mote\_1\_PowerMon.kicad\_sch

**Title: Bus Power Monitor**

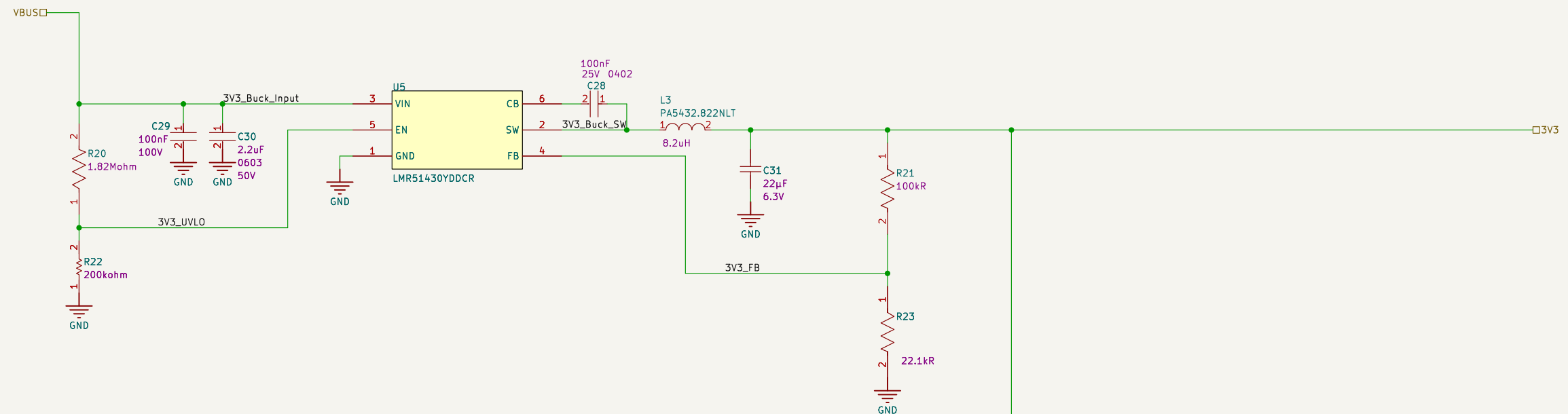
Size: A3 Date: 2026-10-07

KiCad E.D.A. 9.0.6

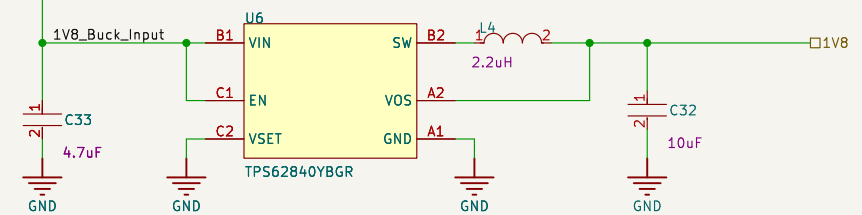
Rev: AA

Id: 6/7

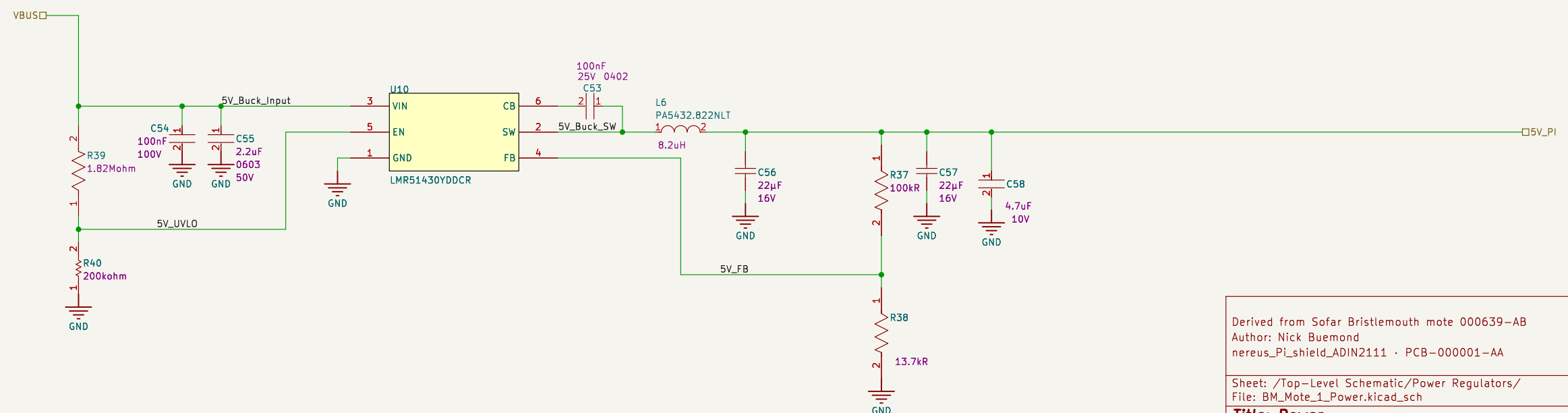
3V3 buck converter: U5 LMR51430, VBUS  $\rightarrow$  3V3



1V8 buck converter: U6 TPS62840, 3V3  $\rightarrow$  1V8 (ADIN AVDD)



5V buck converter for the Pi: U10 LMR51430, VBUS  $\rightarrow$  5V\_PI



Derived from Sofar Bristlemouth mote 000639-AB  
Author: Nick Buemond  
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Sheet: /Top-Level Schematic/Power Regulators/  
File: BM\_Mote\_1\_Power.kicad\_sch

**Title: Power**

|                    |                  |
|--------------------|------------------|
| Size: A3           | Date: 2026-10-07 |
| KiCad E.D.A. 9.0.6 |                  |

Rev: AA  
Id: 7/7